



Confined ionic liquids in covalent organic frameworks toward the rational design of high-safety lithium metal battery

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ABSTRACT

To solve the serious safety issues such as highly flammable organic electrolyte, this work developed a novel lithium metal battery containing quasi-solid-state electrolyte (QSSE) with ionic liquid (ILs)/Li salt as electrolyte and covalent organic frameworks (COFs) as hosts. The micro-scale structure and dynamics properties of several ILs confined in COFs were investigated in detail using molecular dynamics (MD) simulations. It is found that the distinctly alternate layering structure of cations and anions of ILs from the pore surface to pore center are formed inside COFs due to the strong interaction between cations/anions and pore wall of COFs. The orientational preferences of confined ILs are observed depending on the size of anions. Inside the larger pores of COFs, the bulk-like structure of confined ILs is found, which tends to enlarge the leakage risk of electrolytes, while the smaller pore size of COFs can reduce the diffusion of ILs and further limit the ionic conductivity of electrolytes. The prepared QSSE with the appropriate-size ILs confined in medium pore-size COFs shows a higher ionic conductivity (0.87 mS cm^{-1} at 30°C), more efficient inhibition of lithium dendrite (above 2000 h stability) and better battery performance (150.4 mAh g^{-1} at 0.1C). These findings from both simulations and experiments confirm the potential of COFs for achieving nondendritic lithium growth in lithium metal batteries.

1. Introduction

Lithium metal batteries (LMBs) have been widely considered as the next generation of energy storage devices owing to the high theoretical capacity (3860 mAh g^{-1}) and low electrode potential (-3.045 V vs. Standard Hydrogen Electrode) of lithium metal anode [1,2]. However, safety issues caused by highly flammable conventional organic electrolyte and the growth of lithium dendrite impedes their practical applications. Under these circumstances, numerous attempts have been devoted to inhibit electrolyte combustion and the formation of lithium dendrite, including additives [3], lithium metal protection [4], and separator modification [5]. Unfortunately, the lithium dendrite issue still remains little improvements since the unstable and fragile solid electrolyte interphase (SEI) breaks during charge and discharge, leading the lithium metal electrode to be re-exposed to the electrolyte. Alternatively, quasi-solid-state electrolytes (QSSEs) containing ionic liquid electrolytes (ILEs) and a robust skeleton provide a promising solution for

high-safety LMBs, because the introduction of ILEs alleviates the safety problems of conventional organic electrolyte under thermal runaway conditions, and solid matrix with high mechanical strength can inhibit the growth of lithium dendrite.

Currently, lots of research work focus on conventional polymer matrixes, such as poly(ethylene oxide) (PEO) [6], poly(vinylidene fluoride) (PVDF) [7], and poly(methyl methacrylate) (PMMA) [8]. However, their mechanical properties deteriorate when these polymers are immersed in ILs, especially at high temperatures. Recently, a new class of QSSEs using permanent robust matrix instead of polymer skeleton has attracted attention due to the improvement of inhibition of lithium dendrite growth [9,10]. Covalent organic frameworks (COFs), construed from small organic molecules (including boroxines, triazines, imines, and hydrazones) via covalent bond [11,12], are the novel crystalline. COFs are rigid and highly porous materials, and usually possess abundant one-dimensional (1D) regular channels, high specific surface area, and rich adsorption sites [13–15], which can provide better

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mechanical properties in comparison to conventional polymers as the host for QSSEs containing ILEs. Although several IL@COF systems have been investigated [16–21], the reports related to the electrochemical properties, especially ionic conductivity, of ILEs encapsulated in COFs are still insufficient. From previous work, the ionic conductivity of ILEs under confinement in porous materials mainly depends on the content of ILEs, pore size of host, and the interaction between ILEs and hosts [22–24]. In general, the ionic conductivity of the QSSEs containing confined ILEs is usually enhanced with the increase of ILEs content and pore size of the porous materials. However, excess ILEs content and large pore sizes of the host lead to the leakage risk of the QSSEs, which leads to safety problems such as corrosion and environmental pollution. Moreover, the interaction between ILEs and the walls of host has a significant effect on the adsorption capacity and transport properties of confined ILEs. From these perspectives, the fundamental understanding about the structural and dynamic properties of ILEs confined in the COFs with different pore sizes and types is of great interests to the screening and rational design of the novel COFs@ILEs as the ideal QSSE for lithium metal batteries.

Considering the high designability of ILEs and COFs, the combination of COF@ILEs is nearly infinite. Thus, the screening or design of proper COFs@ILEs using the traditional experimental methods is time-consuming and resource-intensive. Fortunately, molecular dynamic (MD) simulation methods have been extensively utilized to investigate the structure and transport properties of bulk and confined ILEs at a molecular scale [25–31]. For example, Li et al. [26] studied the difference of the dynamic and structural properties of imidazolium based IL ([C₄mim][Tf₂N]) confined in silica and carbon mesopores using MD simulations. The calculated diffusion coefficients of cation are consistent with those obtained by nuclear magnetic resonance (NMR) experiments. Particularly, the diffusion properties of ILEs confined in COFs and polymeric ionic liquid (PIL) were calculated by MD simulations in our

previous work [32], and the confined ILEs in COFs were found to transport faster than that in PIL. Based on this, the PIL based composite QSSE with COFs as the additives to enhance the diffusion of ILEs was proposed and presents better electrochemical performance than QSSE without COFs additives, which further motivates us to probe the novel and high-safety QSSE consisting solely of the COFs as host and ILEs. Through MD simulations, the effects of pore size, interfacial functional groups of COFs, and the COFs/ILEs interaction on the micro-scale structure and dynamics properties of COFs@ILEs systems are expected to be revealed to predict the ionic conductivity of the combination of COFs and ILEs for the high-safety QSSE.

In this work, the micro-scale structure and dynamic properties of ILEs containing different anions confined in seven common borate and imine COFs were systematically studied by MD simulations. The best design scheme of COF@ILE system as QSSE was proposed. The Imine based QSSEs were successfully prepared and their ionic conductivity were measured. The highest ionic conductivity (2.0 mS cm^{-1} at 60°C) was achieved for imine based QSSE.

2. Computational and experimental section

2.1. MD simulation details

The structure models of COFs and ILEs for MD simulations are shown in Fig. 1. Imine-1, Imine-2, Borate-1, Borate-2, and Borate-3 represent COFs with 1.7 nm, 3.3 nm, 2.7 nm, 3.2 nm, and 4.7 nm hexagonal pores in diameters, respectively. Imine-2, Imine-3, and Imine-4 have similar pore sizes with different surface groups, such as hydroxyl and methoxy groups. The single COF was stacked layer by layer in a cube box with the gap about 3–4 Å to generate the COFs models. Afterwards, the COF models were filled with a certain amount of ILEs using Packmol software

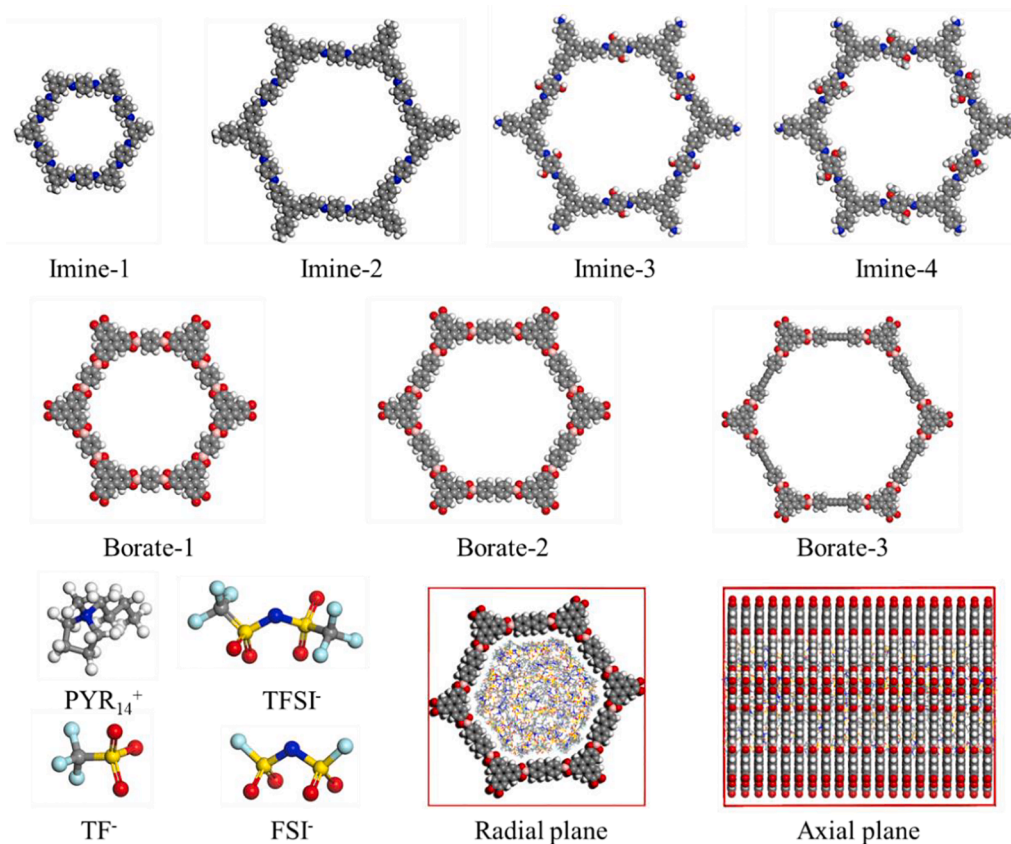


Fig. 1. Structure models of COFs and ILEs for MD simulations (gray: C, white: H, dark blue: N, red: O, yellow: S, light blue: F, pink: B). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

[33], as shown in Fig. 1. The number of molecules in the pores was determined according to the free volume obtained by Material Studio 5.0 (Table S1) [34]. The Lennard-Jones parameters of COFs were taken from CVFF force field [35] and the atomic charges of COFs were obtained by Dmol3 Model at GGA-PBE functional. The parameters of ILs, including Lennard-Jones term, bond stretching, angle bending, dihedral angle term, and coulomb term, were adopted directly from the previous work [36,37]. All the production simulations were conducted under canonical (NVT) ensemble using Gromacs 5.0 package [38] with a time step of 1 fs. The temperature was controlled at 333 K using Hoover-Nose thermostat. The long-range electrostatic interactions with a cutoff of 1.2 nm were applied using the Particle Mesh Ewald (PME) method. Three directions of periodic boundary conditions (PBC) were used. The initial atomic velocities were generated according to a Maxwellian distribution with the LINCS algorithm to constrain all covalent bonds regarding hydrogen atoms. It should be noted that the COFs were fixed throughout the simulations. The density probability distribution of the cations and anions of ILs was performed by MD Analysis package [39] at Python 3.6 environment. The translational diffusion coefficient was calculated from the slope of mean square displacement (MSD) according to the Einstein relation.

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} \langle [r_i(t) - r_i(0)]^2 \rangle \quad (1)$$

2.2. Synthesis and characterization

2.2.1. Material

1-butyl-1-methyl-pyrrolidinium bis(trifluoromethylsulfonyl)imide ($\geq 98\%$, Aladdin, $\text{PYR}_{14}\text{TFSI}$) was dried under vacuum at 80°C for 48 h and stored in glove box before used, 1,3,5-tris(4-aminophenyl)benzene ($\geq 95\%$, Alpha), 1,4-benzenediamine ($\geq 97\%$, Aladdin), 1,4-phthalaldehyde ($\geq 98\%$, Alpha), 1,3,5-benzenetricarboxaldehyde ($\geq 98\%$, Alpha), 2,5-dihydroxyterephthalaldehyde ($\geq 96\%$, Alpha), 1,4-benzenedicarboxaldehyde ($\geq 98\%$, Alpha), 1,4-dioxane ($\geq 98\%$, Aladdin), mesitylene ($\geq 99.5\%$, Macklin), tetrahydrofuran ($\geq 98\%$, Aladdin) and n-hexane ($\geq 98\%$, Aladdin) were used without further purification.

2.2.2. Preparation of COF based QSSEs

The imine-linked COFs were prepared according to previous work [32,40]. For example, the Imine-1 was synthesized by a condensation reaction of 1,3-benzenediamine (BD) and 1,3,5-benzenetricarboxaldehyde (BT). First, 32.4 mg of BD and 32.4 mg of BT were dissolved in a 5 mL of 1,4-dioxane/mesitylene (v: v = 4:1) mixture solution in a pressure bottle reactor under inert atmosphere. After the solid was dissolved completely, 0.5 mL of acetic acid solution (6 M) was added to the reactor. Then the mixture solution was heated at 120°C for 72 h. Finally, the precipitate was collected via centrifugation, washed by Soxhlet extraction method using tetrahydrofuran and n-hexane in sequence. Other COFs were prepared by the same method mentioned above.

In order to ensure continuous lithium ion transport, ionic liquid electrolytes (ILE) with the molality of 0.8 mol kg^{-1} was prepared by dissolving LiTFSI (0.23 g) into $\text{PYR}_{14}\text{TFSI}$ (1 g) in an argon-filled glove box. The COFs based QSSEs were prepared by vacuum assisted capillary process. First, the ILE and COFs with a certain mass ratio were mixed in a pressure bottle by physical stirring. The mixture was then placed in a vacuum oven at 100°C for 48 h. After cooling, the product was stored in an argon-filled glove. Finally, the QSSE was pressed into pellet by a 12 mm diameter mold.

2.2.3. Characterization details

FT-IR spectrometer (PerkinElmer, Spectrum 100), 500 MHz NMR spectrometer (Bruker, AVANCEIII) and X-ray diffraction (Bruker, D8 Advance) were performed to confirmed the structures of COFs. Morphological investigation of the powder samples was obtained using TEM (Talor, F200X). The surface and cross section morphological

investigations of the COF based QSSE were examined from SEM (FEI, NanoSEM 450). The elemental composition of QSSE is determined by an Elemental analyzer (Elementar, VARIO ELIII). The interface between lithium metal and QSSE was investigated by X-ray photoelectron spectroscopy (XPS, Escalab 250Xi, Thermo Fisher). N_2 adsorption/desorption isotherms were performed by automated gas sorption analyzer (Autosorb iQ). The Young's modulus of COFs film was characterized by and atomic force microscope (AFM, Bruker Dimension ICON).

The ionic conductivity of COF based QSSEs was measured by AC impedance spectroscopy measurement using a CHI660D electrochemical workstation in the frequency range 0.1 Hz–1 MHz. Before measurement, the QSSE was sandwiched between two stainless steel electrodes, then sealed in a Swagelok-type cell. The temperature was controlled by the oven and balanced for at least half an hour. The ionic conductivity (σ) of all samples was calculated using the following equation

$$\sigma = \frac{L}{RS} \quad (2)$$

where L is the thickness of sample, R is resistance value of the bulk electrolyte measured by AC impedance spectroscopy, and S is the area of the stainless-steel electrode. The linear sweep voltammetry (LSV) experiments were performed at a scanning rate of 1.0 mV s^{-1} from 2.4 V to 6 V (vs. Li^+/Li). Cyclic voltammetry (CV) behavior was obtained in the voltage ranges of 2.4–5 V at a scanning rate of 1 mV s^{-1} . The galvanostatic tests of Li/Li symmetric cell and charge and discharge tests of $\text{LiFePO}_4/\text{QSSE}/\text{Li}$ cell were conducted using a LANDHE CT2001A charge/discharge system.

3. Results and discussion

3.1. Layering structures of confined ILs in COFs

It is generally believed that the structure features of the confined ILs in the pore of COFs have an important influence on the physical and chemical properties of the COFs/ILs hybrid systems. Initially, the density probability distribution of the cations and anions of $\text{PYR}_{14}\text{TFSI}$ confined in the different COFs nanoscale pore from MD simulations are shown in Fig. 2 and Figure S1, respectively. It is apparent that the density probability distributions of the anion (Fig. 2a–2d) and cation (Fig. 2e–2h) show a distinctly alternate layering structure from the pore surface to pore center. This phenomenon suggests that the coulombic ordering of cations and anions of ILs in bulk systems is rearranged due to the geometrical restriction in 1D channel of COFs and the likelihood of strong interaction between ILs and the walls of the matrix pores, well consistent with previous work [41]. In particular, the size of the hexagonal pores has a significant effect on the arrangement structure of ILs. For example, in the pore of Imine-1 with the smallest channel (or pore size), the cations and anions of ILs are mainly distributed closer to the pore surface of COFs due to the limited pore space of Imine-1. In medium pores, such as Borate-1, Borate-2, and Imine-2 (Figure S1e), the density-rich domains of cations and anions are more obviously separated with layering behaviors. In the biggest pore (Borete-3), the bulk-like structure of ILs is observed at the center of the channel, except two layers of anions/cations enrichment areas close to the wall of pore. The $\text{PYR}_{14}\text{FSI}$ and PYR_{14}TF confined in Imine-2 and $\text{PYR}_{14}\text{TFSI}$ confined in Imine-3 and Imine-4 were also investigated as shown in Figure S1. From the Figure S1a–S1b, it is obvious that the bulk-like structures of confined $\text{PYR}_{14}\text{FSI}$ and PYR_{14}TF in Imine-2 can be observed owing to the smaller size of FSI^- and TF^- anion.

In order to further gain insight into the effect of pore structure on cations and anions arrangement, the quantitative number densities of cations and anions of ILs as a function of pore radius are exhibited in Fig. 3a–3d and Figure S1k–S1o. As depicted in Fig. 3a and Fig. 3b, it is clearly seen that the number density profiles of cation and anion

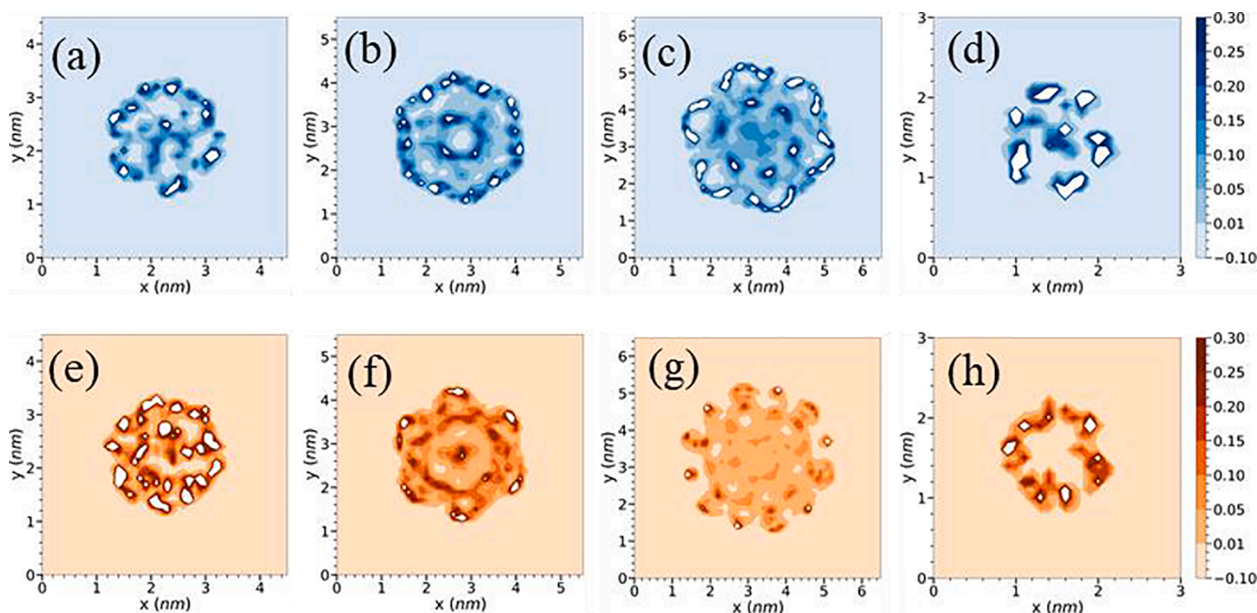


Fig. 2. Probability distribution of the anion and cation of the PYR₁₄TFSI confined in Borate-1(a, e), Borate-2(b, f), Borate-3(c, g), and Imine-1(d, h).

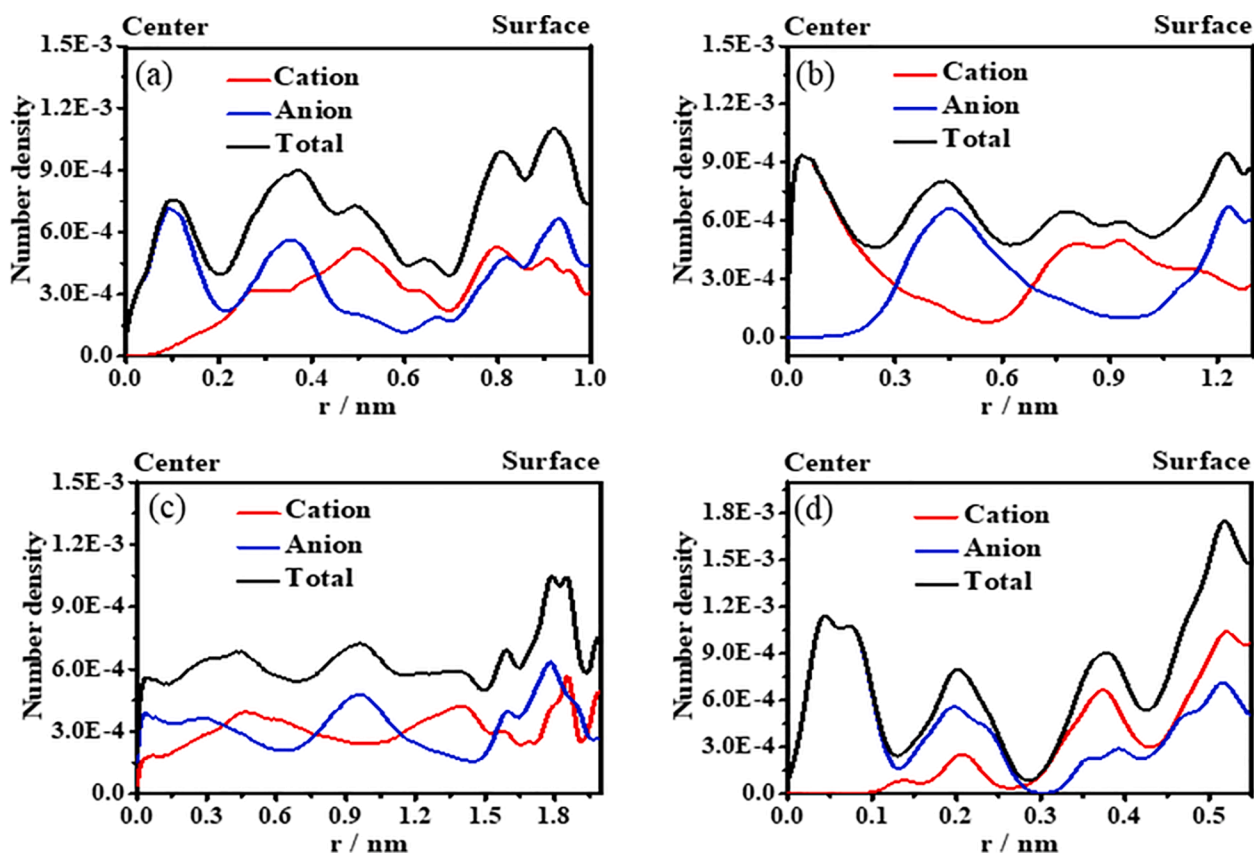


Fig. 3. Number densities of the anions and cations of the PYR₁₄TFSI confined in Borate-1(a), Borate-2(b), Borate-3(c), Imine-1(d).

fluctuate sharply and periodically, in good agreement with the results of density probability distributions. Moreover, the density of anion close to the pore surface is higher than that of cation, which is attributed to the stronger interaction between anions and abundant benzene rings on pore walls. Nevertheless, the different trends are found in both Imine-1 and Borate-3. For Imine-1, the density profiles of cation and anion vary synchronously with the density of anion close to the pore surface lower

than that of cation. This result is probably because that the limitation of pore space leads to the accumulation of flexible anions in the center of pore, thus reducing the density of anions near the wall. In the surface of Borate-3, the densities of cation and anion are nearly equal and the fluctuation trend is similar to that of the bulk system (Figure S2). The former is mainly ascribed to the introduction of triple bond, an electron-rich group with strong attraction to cations, at the edge of the hexagonal

pore, while the latter verifies that the ILs in the center of Borate-3 is beyond the interaction range of the surface groups. This phenomenon has been reported in many mesoporous materials, such as porous carbon, and porous silica ($d \geq 6$ nm). The central ILs maintaining its bulk properties tend to escape from the hybrid system, resulting in leakage risk, while the incorporation of viscous ILs into microporous (≤ 2 nm) usually require vacuum-assisted capillarity and high temperature. Therefore, in this work, medium-aperture (~ 3 nm) COFs, such as Imine-2, Borate-1, and Borate-2, are considered to be the optimal solution to preparation of QSSE for energy storage.

Moreover, the number density of cation and anion close to the pore surface is determined by the type of ILs (Figure S1k-S1l) and surface groups (Figure S1m-S1o) of COFs matrices. From the Figure S1k-S1l, the number density of TF^- anion near the pore surface is significantly higher than that of PYR_{14}^+ cation, while the number density of FSI^- anion is similar to that of cations, indicating that the interaction between FSI^- anion and pore wall is weaker than that between TF^- anion and pore wall. As shown in Figure S1m-S1o, the introduction of hydroxyl and methoxy groups on the pore wall causes a sharp decrease in the number density of ILs close to the pore wall, which is mainly due to the decrease of the free volume with the occupation of functional groups. In addition, the number density of cation is obviously higher than that of anion in Imine-3, while the situation is opposite in Imine-4.

3.2. Orientational structures of ILs confined in COFs

The interfacial structure between ILs and COFs is of great significance for the design and synthesis of QSSEs with specific functions. Thus, the orientation profiles of FSI^- , TF^- , and TFSI^- , represented by N-F vector, C-S vector, and S1-S2 vector, respectively, as a function of the distance from the pore center to pore surface are shown in Fig. 4. From Fig. 4a, there are symmetrically distributed high probability regions of N-F vector with the Cosine value approaching ± 1 , which means that the N-F vector tends to point towards the pore surface. This is probably due to the formation of hydrogen bond between the highly electronegative fluorine atoms and the hydrogen atoms on the pore surface. The probability distribution of C-S vector with only a negative value in the first sublayer and the extremely higher probability with the Cosine value close to -1 are observed, indicating that the S-C vector presents stronger perpendicular orientation toward the pore wall (Fig. 4b). It also suggests that fluorine atoms have a stronger affinity for the interface between ILs and COFs than oxygen atoms and a similar finding is found on the surface of SiH_2 surface [42]. However, for S1-S2 vector of TFSI^- anion, the

high probability regions exist in both vertical and parallel axial direction, which is attributed to its long and flexible molecular structure (Fig. 4c). The layer space of the COF models is 3–4 Å, so the head of the TFSI^- anion may be adsorbed on the upper layer and the tail on the lower layer, resulting in the S1-S2 vector parallel to the axial direction. To sum up, the distinct orientational preferences of anions in this work provide a strong basis for understanding the adsorption mechanism of ILs in COFs.

3.3. Dynamics of confined ILs in COFs

Understanding the dynamic properties of confined ILs plays an important role in predicting the ionic conductivity of these hybrid systems. As shown in Fig. 5, the MSD in a long time range of 30 ns was used to describe the translational behaviors of confined $\text{PYR}_{14}\text{TFSI}$ in different types of COF pores, and the diffusion coefficients were calculated according to the Einstein relation, as listed in Table 1. Initially, the diffusion of cation is faster than that of anion, in agreement with both experiments and simulations [43,44]. It is clear that the MSD profiles of the cation and anion confined in the pore are much lower than that in the bulk system, suggesting that the geometric constraint gives a rise to slower movement. This finding is in good agreement with the results of ILs confined in carbon and silica pore in the work of Li [26]. Owing to the larger pore size, the MSD profile of Broate-3 is similar to the bulk system, which rationalizes the appearance of the bulk phase in the probability distribution shown in Fig. 4c. Moreover, as the pore size decreases from 4.6 to 1.7 nm, the MSD profiles decline dramatically and the diffusion coefficients of cation and anion decrease from 7.1 and 4.7 to 0.14 and 0.12, respectively. Coasne et al. [43] systematically studied the influence of the ILs loading fraction in silica pore on the diffusion coefficient, and found that the diffusion coefficient decreased as the loading fraction reduced. They attributed this observation to the fact that there was only a “stiff” ionic layer on the silica pore surface at low loading. In this work, the “stiff” ionic layer is also found in Imine-1 with the smallest pore size of 1.7 nm, which explains the minimal diffusion coefficient. However, the ILs in the pore center recover its bulk properties as the increase of pore size and the faster diffusion coefficients are achieved. The MSD profiles of Imine-2, Imine-3, and Imine-4 are almost overlap, resulting in the nearly identical diffusion coefficients, except that the translational behavior of cation in Imine-2 is faster than that in Imine-3 and Imine-4. This can be ascribed to the introduction of hydroxyl and methoxy groups containing oxygen atom, which enhances the interaction between the pore surface and cation. In addition, the diffusion coefficients of $\text{PYR}_{14}\text{FSI}$ and PYR_{14}TF confined in COFs were

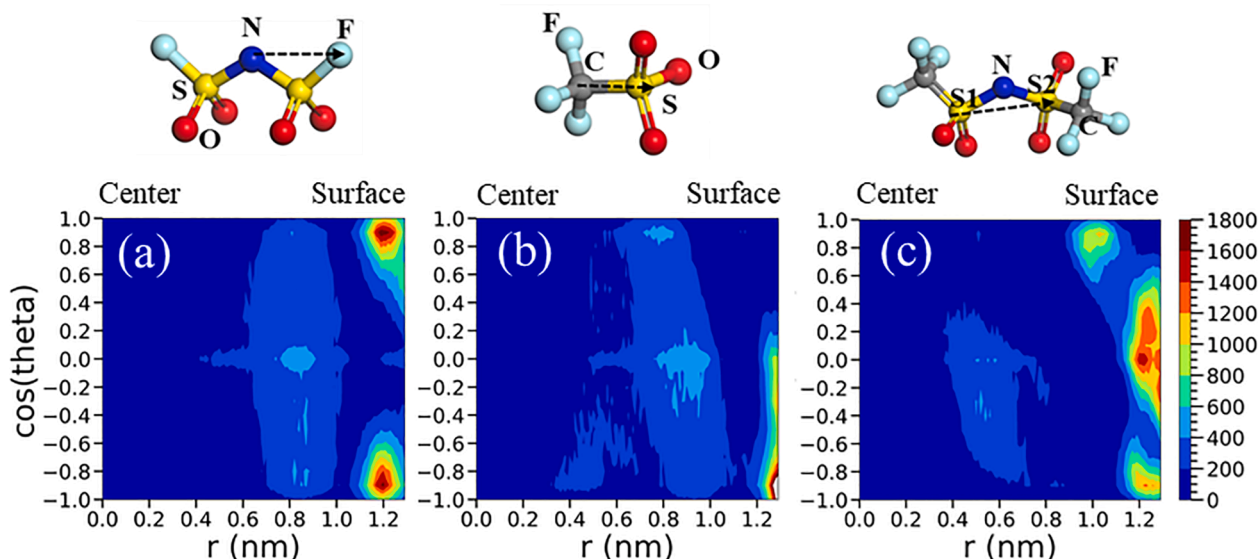


Fig. 4. Orientational distribution of (a) FSI^- , (b) TF^- , and (c) TFSI^- with respect to positive Z-axis in Imine-2.

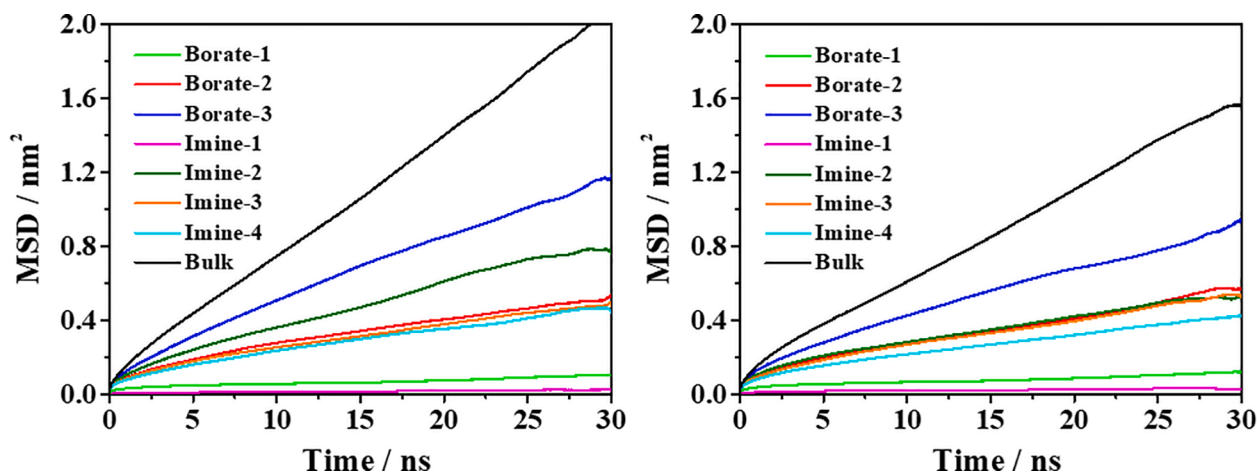


Fig. 5. MSD profiles of cation (left) and anion (right) of PYR₁₄TFSI as a function of time for the hybrid systems. Bulk system is also included.

Table 1

Diffusion coefficients of bulk and confined PYR₁₄TFSI in different COFs.

Sample	Cation/ 10 ⁻¹² m ² s ⁻¹	Anion/ 10 ⁻¹² m ² s ⁻¹	Sample	Cation/ 10 ⁻¹² m ² s ⁻¹	Anion/ 10 ⁻¹² m ² s ⁻¹
Bulk	9.3	6.2	Imine-1	0.14	0.12
Borate-1	0.45	0.36	Imine-2	4.0	2.1
Borate-2	3.5	2.3	Imine-3	3.1	2.3
Borate-3	7.1	4.7	Imine-4	2.9	2.0

also calculated (Table S2 and Table S3), which shows a similar trend.

Based on the above simulation results, it is reasonably inferred that ILs with appropriate molecular size confined in the COFs with medium pore size (forming two ionic sublayers) are required to prepare QSSEs with both moderate ionic conductivity and no leakage risk. To further confirm conjecture, a series of QSSEs based on imine COFs were prepared and successfully applied to lithium metal battery. It should be noted that the important reason for choosing imine COFs is to be that it is generally believed that imide COFs are more stable than borate COFs. However, it is a pity that the synthesis of COFs with an aperture larger than 4 nm is very difficult.

3.4. Novel COFs based QSSEs

The detailed characterization results of the prepared COFs are provided in the Supporting Information, including PXRD, TEM, FT-IR, BET and solid ¹³C NMR (Figure S3-S14). From the PXRD profile, all COFs exhibit obvious crystallization peaks, consistent with previous works [40,45], which indicates that the COFs obtained in this work possess good crystallinity. The TEM images of COFs show that the COFs tend to grow into microspheres with a diameter of several hundred nanometer layer by layer. The formation of C = N bond is proved by the signal of FT-IR (1617 cm⁻¹) and solid ¹³C NMR (110–152 ppm) [46]. From the BET results, the COFs exhibit high BET surface areas of 850–2096 m² g⁻¹ with average pore size of 1.2–3.2 nm and pore volume of 0.73–1.51 cm³ g⁻¹. Except for Imine-1, the pore volume of other COFs is larger than that of MCM-41 (0.78 cm³ g⁻¹) [9]. Therefore, it is believed that the COFs can absorb more ILE and show higher ionic conductivity.

In our previous work, the feasibility of ILs confined into COFs has been confirmed by density functional theory (DFT) and TEM results [32]. Furthermore, the fraction of ILs determines the ionic conductivity of the QSSEs and the difficulty of subsequent processing. Therefore,

Imine-4 based QSSEs with different PYR₁₄TFSI electrolyte contents were prepared by a vacuum-assisted method in this work (Figure S15). In general, with the increase of ILE content, the samples change from solid-like to liquid-like and gradually lose their processability. In order to further investigate the composition of the QSSEs, the elemental analysis was performed and the results were listed in Table S4. It is obvious that the sulfur content of the samples is close to the theoretical value, indicating the ILE is completely encapsulated in the Imine-4. In particular, the sulfur content of sample 8:1 is slightly less than theoretical value, which means that ILE escapes from the hybrid system. As a result, a ratio of 6:1 is considered to be the best choice for Imine-2, Imine-3, and Imine-4 to prepare QSSEs. Notably, as for Imine-2, the morphology becomes liquid-like when the IL electrolyte content is greater than 67 wt%, which is attributed to its smallest pore size.

The COFs based QSSE pellet without binders are successfully prepared by the mold pressing method and the thickness of the sheet is 152.8 μm (Fig. 6a and Fig. 6c). The SEM images of surface and cross section of the QSSE are shown in Fig. 6b and Fig. 6d, respectively. Although the COF particles are tightly bound together, there are still several micron-sized gaps in the interior of QSSE. The ILE transmitted in these gaps should follow the law of the bulk phase electrolyte, which is conducive to the improvement of ionic conductivity. In addition, the

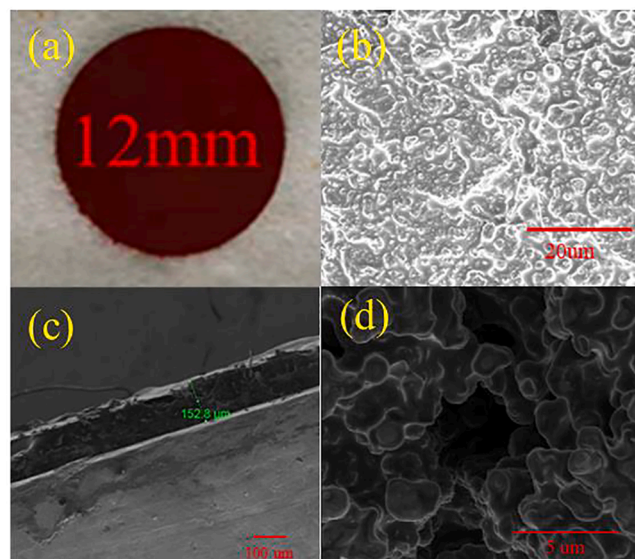


Fig. 6. Photograph of COFs based QSSEs (a) and SEM images of its surface (b) and cross section (c-d).

Young's modulus of the QSSE pellet was measured by AFM method (Figure S16). It can be seen that the sample exhibits uniform modulus distribution and an average Young's modulus of 2.6 Gpa. The high Young's modulus is beneficial to inhibit the growth of lithium dendrites.

3.5. Ionic conductivity of COF based QSSE

Ionic conductivity is the most important parameters of QSSE for electrochemical application. In order to realize the conduction of lithium ions, LiTFSI was introduced into QSSEs to provide lithium source. The ionic conductivity of the COF based QSSEs were measured using AC impedance spectroscopy, and the results are presented in Fig. 7 and Table 2. The original data of Nyquist plots were shown in Figure S17. From Fig. 7, the ionic conductivity increases with increasing temperature and exhibits Arrhenius behavior, which is consistent with previous work [23]. Furthermore, the content of ILE in the COF based QSSE directly determines its ionic conductivity. From Fig. 7a, the Imine-4 based QSSE Imine-4@67, Imine-4@80, and Imine-4@87 show ionic conductivity of $1.4 \times 10^{-5} \text{ S cm}^{-1}$, $1.5 \times 10^{-4} \text{ S cm}^{-1}$, and $5.0 \times 10^{-4} \text{ S cm}^{-1}$ at 30°C with an activation energy of 0.37 eV, 0.35 eV and 0.30 eV, respectively. Combined with the results of MD simulation (Fig. 2 and Fig. 5), the ILE tends to form a single "stiff" ionic layer which shows slowest diffusion coefficient at the lower ILE content. With the increase in ILE content, the second ionic layer gradually forms until the space of pores is fully filled with ILE. Therefore, the ionic conductivity increases with the increase of ILE content, while activation energy is the opposite. It should be noted that the powder shows liquid-like when the ILE content is as high as 89 wt% and cannot be pressed into self-standing pellet. This is due to the overflow of ILE from the pore, which make the QSSE present slurry state. As expected, the highest ionic conductivity is achieved when the ILE contents reach 87 wt%. The highest ionic conductivity is greater than the conductivity of the MCM-41 based QSSE ($4.0 \times 10^{-4} \text{ S cm}^{-1}$) prepared by Han et al. [47]. This improvement is mainly attributed to the higher specific surface area and pore volume of COFs prepared in this work than MCM-41, and thus the COFs based QSSEs have higher ILE content. Additionally, the ionic conductivity of Imine-1@67, Imine-2@87 and Imine-3@87 is $2.5 \times 10^{-4} \text{ S cm}^{-1}$, $8.7 \times 10^{-4} \text{ S cm}^{-1}$, $6.2 \times 10^{-4} \text{ S cm}^{-1}$ at 30°C with an activation energy of 0.26 eV, 0.29 eV, and 0.32 eV, respectively (Fig. 7b). The Imine-1@67 shows the lowest ionic conductivity because of the lowest pore volume and the slowest ionic diffusion shown in Fig. 5. Apparently, the ionic conductivity of COF based QSSE strongly relies on the structure of COF host. When the volume of functional groups on the pore surface is reduced, the ionic conductivity increases, which is probably attributed to faster ionic diffusion shown in MSD results (Fig. 5). Accordingly, the activation energy decreases as the volume of functional groups

Table 2

Ionic conductivity of different QSSEs.

matrix	Ionic liquid	Li-salt	ILE content (wt%)	Ionic conductivity ($10^{-4} \text{ S cm}^{-1}$ at 30°C)	Ref.
Imine-1	[PYR ₁₄][TFSI]	LiTFSI	67	2.5	This work
Imine-2	[PYR ₁₄][TFSI]	LiTFSI	87	8.7	This work
Imine-3	[PYR ₁₄][TFSI]	LiTFSI	87	6.2	This work
Imine-4	[PYR ₁₄][TFSI]	LiTFSI	87	5.0	This work
MCM-41	[PYR ₁₄][TFSI]	LiTFSI	77	4.0	[47]
PEO	[PYR ₁₄][TFSI]	LiTFSI	48	4.4	[48]
PIL	[PYR ₁₄][TFSI]	LiTFSI	72	3.2	[49]
MCM-41	[EMIM][TFSI]	LiTFSI	70	6.4	[9]

decreases. Interestingly, the imine-1@67 shows the smallest activation energy with the reason to be that the ILE contributing to ionic conductivity mainly exists in the micron-sized gaps formed between COFs particles, where diffusion resistance is lower than that of nanoscale pores.

Among all QSSEs containing PYR₁₄TFSI-LiTFSI as electrolyte, Imine-2 obtained in this work has the highest ionic conductivity. This finding is mainly ascribed to the ultra-high ILE content and highly open ionic transport channels of Imine-2@87. In addition, Singh and co-worker [9] synthesized QSSEs containing EMIMTFSI-LiTFSI as electrolyte and ordered mesoporous MCM-41 as host and its ionic conductivity is higher than QSSEs containing PYR₁₄TFSI-LiTFSI as electrolyte. This result indicates that the use of imidazolium IL is expected to further enhance the ionic conductivity. In this study, considering the electrochemical stability of imidazolium ILs is lower than that of pyrrolidinium ILs, PYR₁₄TFSI-LiTFSI is still selected as electrolytes for subsequent electrochemical tests. The lithium-ion transference number (t_{Li^+}) of the COFs based QSSE were measured using the steady-state current method proposed by Bruce and Vincent [50] and details were shown in supporting information (Figure S18 and Table S5). The t_{Li^+} values of Imine-1@67, Imine-2@87, Imine-3@87, and Imine-4@87 are calculated to be 0.15, 0.28, 0.16 and 0.23, respectively. The Imine-2@87 shows the highest t_{Li^+} , which is probably due to the preferential interaction between the anions of IL and the pore surfaces of COFs verified by the results of MD simulations (Figs. 3-4). The lower t_{Li^+} of Imine-3@87 and Imine-4@87 is perhaps due to the abundant oxygen atoms which have strong interaction with lithium ion. Based on the above results, the

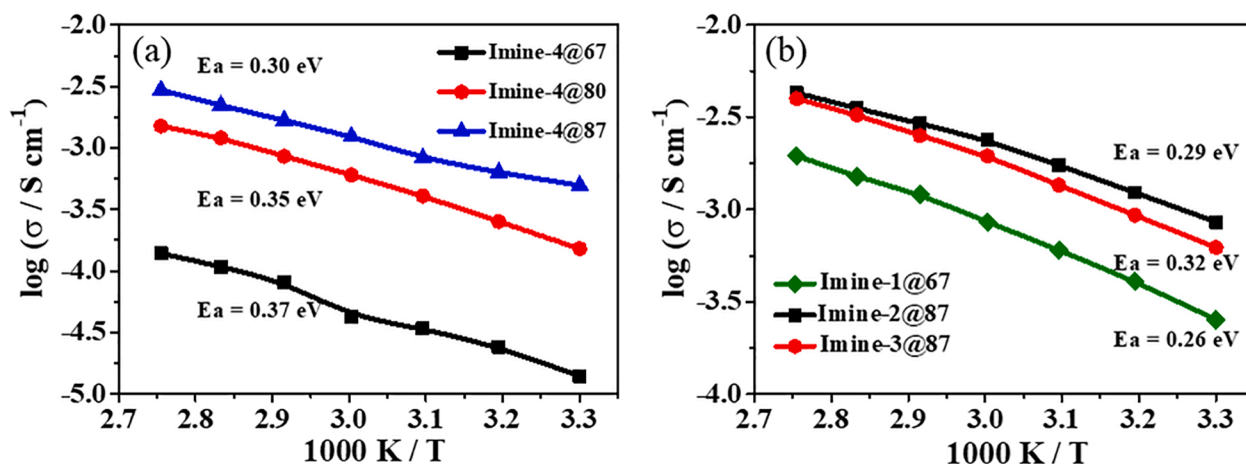


Fig. 7. Temperature dependence of ionic conductivity for the COF based QSSE (a) Imine-4@X (X = mass content of ILE) and (b) the QSSEs with different COFs.

optimized Imine-2@87 with the highest ionic conductivity and t_{Li^+} is selected for further electrochemical characterization.

3.6. Electrochemical performance

To evaluate the stability of COF based QSSE against lithium dendrite formation, the time-dependent voltage curves of lithium metal symmetric cell during galvanostatic charge–discharge test with a current density of 0.1 mA cm^{-2} were shown in Fig. 8. The over potential of the cell with Imine-2@87 increases at first, then decreases, and finally stabilizes at about 76 mV during the 1000 cycles, indicating an even lithium deposition that is able to inhibit unwanted lithium dendrite formation. In comparison, the cell with PEO-Li-IL shows a sudden voltage drop after 680 h cycling (Fig. S19). This may be due to the poor mechanical properties and low lithium ion conductivity of the PEO based QSSE, which leads to the short circuit of the two lithium metal electrodes. In addition, the over potential of the cell with Imine-2@87 at different current densities (0.02 – 0.2 mA cm^{-2}) are also examined and shows no apparent voltage fluctuation (Fig. S20), again indicating interface between lithium surface and QSSE is stable. To further get an insight into the stable interface, the XPS analysis was performed to confirm the existence of LiF, which is probably due to the degradation of anion of ILE (Fig. S21). Moreover, the ability of LiF to stabilize the interface between lithium metal and electrolyte has been reported in many literatures [51–53].

In order to further investigate the inhibition of COF based QSSE on lithium dendrite formation, the surface morphologies of the lithium electrodes after 150 lithium plating/stripping cycles were characterized by SEM. The time-dependent voltage curves of liquid electrolyte show an unstable and increasing voltage, indicating the interface is unstable (Fig. S22). As shown in Fig. 9a, the lithium anode of the cell cycled with liquid electrolyte displays a rough and loose surface with many cracks and bulges. However, lithium anode of the cell cycles with COF based QSSE shows a smooth and shiny surface, furtherly verifying the Imine-2@87 possesses a superior ability to suppress lithium dendrite owing to the high modulus and stable electrolyte/electrode interface (Fig. 9b). Another important parameter for the applications of Lithium metal battery is the electrochemical stability. In this work, linear sweep voltammetry (LSV) and cyclic voltammetry (CV) tests were carried out to investigated the electrochemical stability of COF based QSSE. As shown in Fig. S23, the COF based QSSE exhibits anodic stability up to 5.0 V (vs Li^+/Li), suggesting its potential application in high voltage battery. The CV of $\text{Li}/\text{QSSE}/\text{LiFePO}_4$ cell shows redox peaks at 3.2 V and 3.6 V, corresponding to deintercalation/intercalation of lithium ions into LiFePO_4 . Moreover, the CV curve of the 3rd cycle has the same redox

peaks as that of the first cycle, indicating its excellent reversibility and stability.

Finally, the battery performance of COF based QSSE was evaluated at 60°C , using LiFePO_4 as the cathode material. The mass of active material is about 1.6 mg cm^{-2} . As shown in Fig. 10a, an initial discharge capacity of 163.1 mAh g^{-1} with 96% coulombic efficiency was recorded, and its discharge capacity decreased to 150.5 mAh g^{-1} with 98% coulombic efficiency after 50 cycles. Moreover, the cell showed a discharge capacity of 119.7 mAh g^{-1} with 99% coulombic efficiency after 110 cycles at 0.5C (Figure S24). Fig. 10b displays the charge and discharge profiles of $\text{LiFePO}_4/\text{QSSE}/\text{Li}$ cell at 0.1C ($1\text{C} = 170 \text{ mAh g}^{-1}$) and the cell exhibits a 3.44 V charge plateaus and 3.39 V discharge plateaus during the 50 cycles. The rate performance of $\text{LiFePO}_4/\text{QSSE}/\text{Li}$ cell was further studied with the current rate increasing from 0.1C to 2C, as show in Fig. 10c. The cell shows remarkable rate performance with discharge capacities of 164, 142, 118, 101 and 93 mAh g^{-1} at 0.1C, 0.2C, 0.5C, 1C and 2C rate, respectively. The discharge capacities decrease as the discharge rates increase, which is attributed to migration restriction of lithium ions and the increasing polarizations as suggested by the charge–discharge profiles (Fig. 10d). To sum up, the COF based QSSE obtained in this paper has great potential for application in high performance and high safety lithium metal batteries.

4. Conclusion

In summary, the distribution structure, orientation and diffusion coefficient of three ionic liquid, $\text{PYR}_{14}\text{TFSI}$, $\text{PYR}_{14}\text{FSI}$ and PYR_{14}TF , confined in seven common COFs were systematically studied by MD simulations. The distinctly alternate layering structure of cations and anions of ILs from the pore surface to pore center are formed in the COFs due to the stronger interaction between cations/anions and the pore wall of COFs. The orientational preferences of confined ionic liquid are characterized with different features depending on the size of anion. The bulk-like structure of confined ILs is observed in the larger pore of COFs, which can enlarge the leakage risk of electrolytes, while the smaller pore size of COFs is found to reduce the diffusion of ILs. Hence, it is concluded that the medium-size COFs adsorbing two-layer ionic liquid is suitable for the preparation of high-safety QSSEs with no leakage risk. According to this criterion, a series of QSSEs containing four Imine COFs as host, and $\text{PYR}_{14}\text{TFSI}$ - LiTFSI as electrolyte was prepared. The ionic conductivity of these QSSEs were calculated by AC impedance spectroscopy. Imine-2@87 exhibits the highest ionic conductivity (0.87 mS cm^{-1} at 30°C and 2.0 mS cm^{-1} at 60°C) and is selected for further electrochemical tests. The Li symmetric battery exhibits excellent stability and inhibition of lithium dendrite during the 2000 h cycle. Finally, the

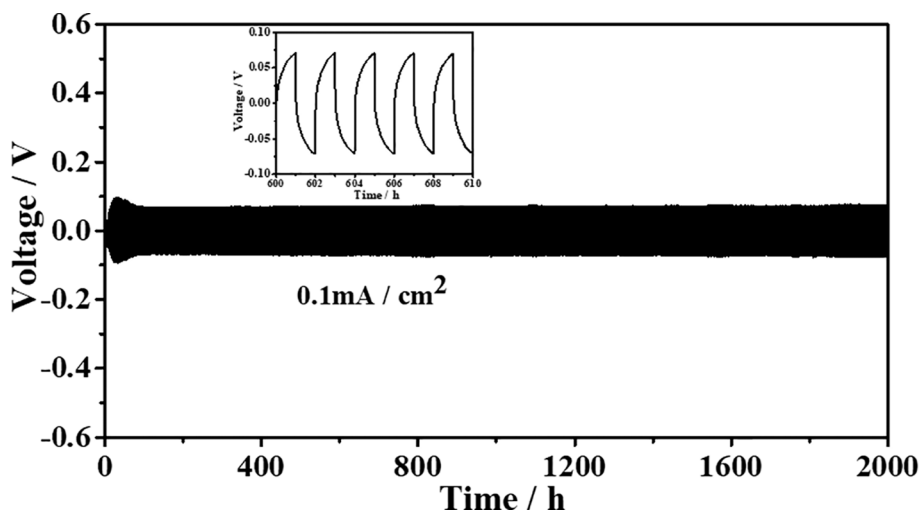


Fig. 8. The voltage–time curves during cycling tests of the $\text{Li}/\text{Imine-2@87}/\text{Li}$ cell at the current density of 0.1 mA cm^{-2} .

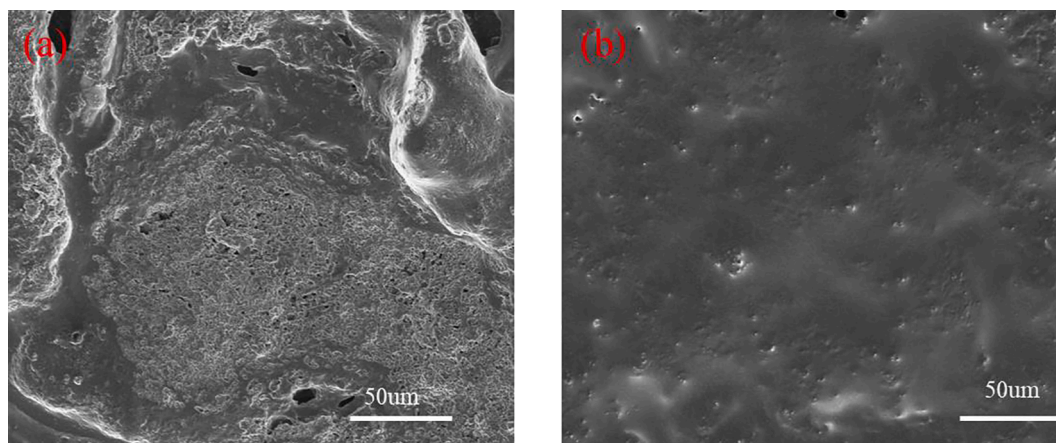


Fig. 9. SEM images of lithium metal surface after 150 lithium plating/stripping cycles. (a) liquid electrolyte (b) COF based QSSE.

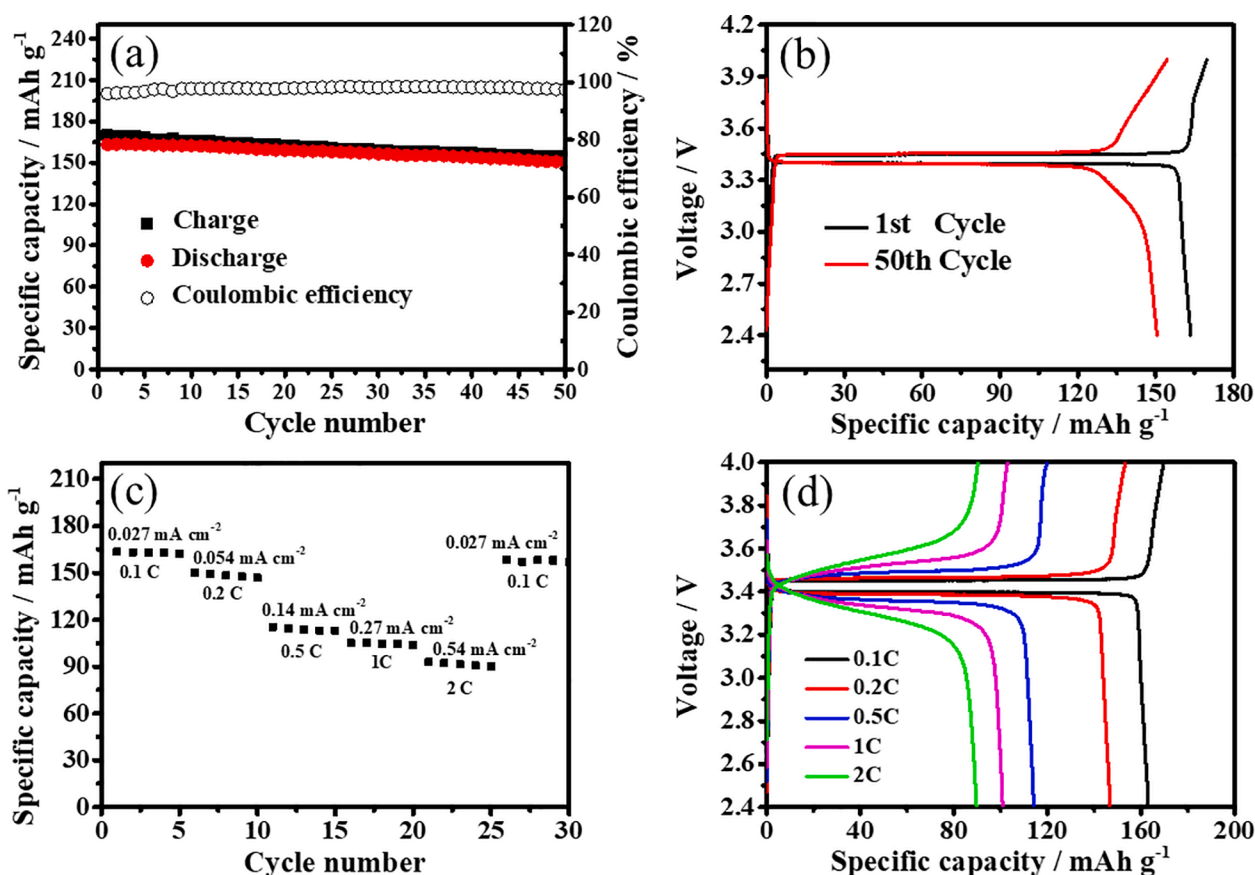


Fig. 10. (a) Charge-discharge capacity and coulombic efficiency as a function of cycle number for LiFePO₄/QSSE/Li cell at 0.1C; (b) Charge-discharge profiles of LiFePO₄/QSSE/Li cell at 0.1C; (c) Rate performance and corresponding charge-discharge profiles of LiFePO₄/QSSE/Li cell.

LiFePO₄/Li cell based on the Imine-2@87 shows an excellent rate performance and delivered discharge capacities of 150.5 mAh /g after 50 Cycles at 0.1C. Hopefully, this work would provide a promising approach to the design of novel QSSEs for high performance and high safety lithium metal batteries.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2021.133749>.

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